

Project Title
Log Analysis System for Recurring Error Patterns

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

The Log Analysis System for Recurring Error Patterns is a DSA-3 project developed to make the process of analyzing large amounts of system and application log data easier and more efficient. Modern applications generate thousands of log entries containing information, warnings, and errors, and checking all these entries manually can take a lot of time. Our system automatically reads the log data, identifies different errors, counts how frequently they occur, and detects errors that keep appearing repeatedly. It also helps in finding similar error messages that may represent the same type of problem. This allows users to quickly understand common issues and focus on errors that require attention.

The project mainly focuses on applying Data Structures and Algorithms to solve this real-world problem. Data structures such as HashMap, ArrayList, HashSet, and Priority Queue are used to store, count, organize, and rank error information. String-processing algorithms such as KMP, Rabin-Karp, and Aho-Corasick are used to search for error patterns efficiently, while Edit Distance using Dynamic Programming helps identify similar error messages. The system finally presents the most frequent and important errors in an organized manner. Through this project, we demonstrate how DSA concepts can be applied to a practical problem to make log analysis faster, more systematic, and easier to understand.

Signature of the Guide